Module 08 Lab Exercise - Contribution Journal

John Grubbs

I completed and ran my own copy of the Module 08 Bias-Variance Tradeoff lab. I created a synthetic sine wave dataset with random noise and plotted the data so I could see the underlying pattern.

I trained polynomial regression models with degrees 1, 4, and 15. I plotted each model's predictions and compared the results. The degree 1 model underfit the data, the degree 4 model provided a good fit, and the degree 15 model overfit the data by creating a very uneven line that followed the individual data points too closely.

I also created a learning curve function using scikit-learn's learning_curve method. I used five-fold cross-validation to compare the training and validation scores for the underfitting and overfitting models. I completed the knowledge check by explaining how the different polynomial degrees represented underfitting, overfitting, and a good fit.

Luis Adrian

My contribution included completing the code to train and plot polynomial regression models with degrees 1, 4, and 15. I used pipelines with PolynomialFeatures and LinearRegression, fitted the models, and generated predictions to compare their complexity against the synthetic data.

I also completed the model definitions and function calls for the learning curves. For the written analysis, I explained underfitting, overfitting, and the relationship between training and cross-validation scores. These tasks helped me connect the programming steps with the goal of building models that generalize to new data.

Keira Burdine

I independently completed the coding activities in Google Colab. I generated synthetic data, created polynomial regression models with different degrees of complexity and plotted learning curves to examine training and cross-validation performance.

During the assignment, I encountered several coding errors the required troubleshooting, particularly with Python syntax and indentation. I corrected these issues, reran the code and verified that the graphs displayed properly. I completed

required activities and documented my understanding of the bias-variance tradeoff through the lab questions and reflection.